



SIMATS Online Programming Lab

JAVA PROGRAMMING

 Akileshwaran A
192511354RD

QUESTIONS

8 / 8 completed

Pro-Consume

PRO-CONSUME

Write a Java program to create a producer-consumer scenario using multiple threads. The producer thread should add items to a queue, and the consumer thread should remove items from the queue and process them.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.LinkedList;
2 import java.util.Queue;
3
4 public class Main {
5
6     // Shared buffer to facilitate communication between Producer and Consumer
7     static class SharedBuffer {
8         private final Queue<Integer> queue = new LinkedList<>();
9         private final int CAPACITY;
10
11         SharedBuffer(int capacity) {
12             this.CAPACITY = capacity;
13         }
14
15         // Method for producer to add items
16         public synchronized void produce(int value) throws InterruptedException {
17             while (queue.size() == CAPACITY) {
18                 wait(); // Wait if the buffer is full
19             }
20             queue.add(value);
21
```

OUTPUT

Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Produced: 4
Consumed: 3
Produced: 5
Consumed: 4
Produced: 6

**INPUT**

Your INPUT goes here!

